MIL-S-475D
23 June 1970
SUPERSEDING
MIL-G-475C
13 April 1966

MILITARY SPECIFICATION

* SUNGLASSES (WITH CASE)

This specification is mandatory for use by all Departments and Agencies of The Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for one type of sunglasses and case.

* 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| L-P-349 | - | Plastic Molding and Extrusion Material, Cellulose Acetate Butyrate. |
| L-P-375 | - | Plastic Film, Flexible, Vinyl Chloride. |
| L-P-504 | - | Plastic Sheet and Film, Cellulose Acetate. |
| QQ-B-613 | - | Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet, and Strip). |
| QQ-B-626 | - | Brass, Leaded and Non-Leaded: Rod, Shaped, Forgings, and Flat Products with Finished Edges (Bar and Strip). |
| QQ-C-586 | - | Copper-Nickel-Zinc Alloy; Rod, Shapes, and Flat Products with Finished Edges (Flat Wire, Strip and Bar). |
| QQ-S-698 | - | Steel, Sheet and Strip, Low-Carbon. |
| QQ-W-321 | - | Wire, Copper Alloy. |
| QQ-Z-325 | - | Zinc Coating, Electrodeposited, Requirements for. |

FSC 8465

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TT-W-572 - Wood-Preservative; Water-Repellent.
UU-P-405 - Paper, Newsprint.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing.

MILITARY

MIL-F-495 - Finish, Chemical, Black, For Copper Alloys.
MIL-P-22748 - Plastic, Polyethylene and Copolymers, High Density, Molding and Extrusion Material.

STANDARDS

FEDERAL

FED-STD-406 - Plastics: Methods of Testing.
FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.

DRAWINGS

ARMY NATICK LABORATORIES

2-1-54 - Glasses, Sun, M-1944, With Case.

(Miniature copy of Drawing 2-1-54 identified as figure 1 is attached for information purposes only).

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).



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* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, N.W., Washington, D. C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

## 3. REQUIREMENTS

3.1 Bid sample approval.- Sunglasses in conformance with Drawing 2-1-54 are desired, however, bidders may offer alternate designs, shapes, and dimensions for the frame, temple and lens components only. All other requirements shall comply with this specification. The bidder shall submit, with his bid, three (3) sunglasses representing the item he proposes to bid on and furnish to the Government if he receives award of the contract. These samples shall be evaluated for conformance of the item with design and construction requirements (see 6.2). Upon approval and award of contracts, these samples shall be used as standards for all design, construction and dimensional characteristics (materials requirements excepted).

3.2 Materials.-

3.2.1 Frame.-

3.2.1.1 Brow bar and nose bridge.- The brow bar and nose bridge shall be injection molded, opaque black, virgin cellulose acetate butyrate conforming to type III, class MH of L-P-349. Fifteen percent by weight of reground cellulose acetate butyrate will be allowed, provided the reground material is the same composition as the virgin material and meets the physical requirements of the virgin material.



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3.2.1.2 Temple (butt).- The temple butt shall be injection molded, opaque black, virgin cellulose acetate butyrate conforming to type III, class MH of L-P-349. Fifteen percent by weight of reground cellulose acetate butyrate will be allowed, provided the reground material is the same composition as the virgin material and meets the physical requirements of the virgin material.

3.2.1.3 Temple (tubing).- The tubing shall be extruded from clear cellulose acetate butyrate material and shall conform to type III, class MH of L-P-349.

3.2.1.4 Hardware.- All hardware, except snap fastener for carrying case, shall be either brass or nickel-silver and shall posses physical properties as indicated in the applicable specifications unless otherwise indicated.

3.2.1.4.1 Hinge (temple).- The temple hinge shall be composition 268 or 260, quarter or half hard temper conforming to QQ-B-613 or QQ-B-626; or composition 745, quarter or half hard temper conforming to QQ-C-586. If composition 745, quarter or half hard temper is used, the nickel content may range between 9.0 and 16.5 percent provided the zinc content only is proportionately reduced as the nickel content is increased.

3.2.1.4.2 Temple, butt and cable (winding and core).- The temple butt and cable (winding and core) shall be composition 745, quarter or half hard temper conforming to QQ-C-586, however, the nickel content may range between 9.0 and 16.5 percent, provided the zinc content only is proportionately reduced as the nickel content is increased.

3.2.1.4.3 Rivet (hinge).- The rivet shall be alloy No. 260, quarter hard conforming to QQ-W-321.

3.2.2 Lens (plastic).- The plastic sheet used for the lenses shall be cellulose acetate butyrate, neutral gray in color.

3.2.2.1 Cellulose acetate butyrate.- The cellulose acetate butyrate shall have a high polish finish and be of optical quality, slow burning, uniform and of construction and shade of color necessary to produce lenses meeting the requirements of this specification. The cellulose acetate butyrate shall meet the optical requirements specified in L-P-504 for type III material except that the luminous transmittance shall be 15 percent to 30 percent.

3.2.3 Carrying case.-

3.2.3.1 Covering.- The covering shall be flexible vinyl chloride film conforming to the requirements for the 0.020 inch thickness of type I, class 2 of L-P-375 with the following exceptions:



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| | |
|---|---|
| Thickness | 0.035 ± 0.003 inch |
| Ultimate elongation, transverse direction | 200 percent (minimum) |
| Stiffness, Clark, at 75°F | 20 ± 3 centimeters |

3.2.3.2 Fastener, snap.- The snap fastener for the carrying case shall be a brass, 16 line segma dot snap fastener consisting of the following component parts, as applicable:

Screw stud (for wood nose block only) - United Carr No. X5-21305 or equal.
Socket - United Carr No. BS-21201 or equal.
Button - United Carr No. BS-21154 or equal.
Stud (for plastic nose block only) - United Carr No. BS-21316 or equal.
Eyelet (for plastic nose block only) - United Carr No. BS 21401 or equal.

The snap fastener shall be finished conforming to MIL-F-495.

3.2.3.3 Nose block.- The nose block shall be either wood or plastic and shall conform to the requirements as specified herein and to dimensions in Drawing 2-1-54.

3.2.3.3.1 Plastic.- The nose block shall be high density polyethylene conforming to class A, grade 1 of MIL-P-22748. The color shall be natural.

3.2.3.3.2 Wood.- The nose block shall be any sound wood, free from holes, shakes, decay or chips, suitable for the purpose intended and shall have 2 coats of wood sealer conforming to composition A of TT-W-572.

3.2.3.4 Reinforcement.- The reinforcement shall be fabricated from steel sheet or strip conforming to commercial quality No. 3 or No. 4 temper of QQ-S-698. The reinforcement shall be finished conforming to type I, class 2 or class 3 of QQ-Z-325.

3.3 Design and construction.-

3.3.1 Frame.- The frame shall conform to the design, shape and dimensions shown on Drawing 2-1-54 within the limits of the standard samples (see 3.1 and 6.2). The design shall be such as to accommodate the plastic lens as shown, provide binocular vision and hold the inner surface of the lens away from the eyes a sufficient distance for adequate eyelash clearance.

3.3.2 Temples.- The vertical and horizontal angles of the two temples as shown on Drawing 2-1-54 within the limits of the standard samples (see 3.1



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and 6.2), shall be matched and shall provide a satisfactory fit at the mitered joint with the brow bar.

3.3.3 Temple butt and tubing (plastic).- The plastic temple butt and tubing shall be securely and permanently cemented together. The tubing shall be permanently heat sealed on the end.

3.3.4 Hinge.- The mating barrels of the hinges shall form a snug fit and shall be held together with a rivet long enough so that there is no appreciable looseness which permits angular up and down movement of the temple with respect to the front. The rivets shall be permanently and securely set. The hinges shall be securely spot welded, soldered or securely riveted to the temple butt insert.

3.3.5 Brow bar and nose bridge.- The brow bar and nose bridge shall be securely cemented to the lens.

3.3.6 Lens.- The lens shall be blanked or cut to shape, size and dimensions shown in Drawing 2-1-54 (thickness 0.035 $\pm$ 0.005) within the limits of the standard samples (see 3.1 and 6.2). The lens shall be such that when inserted in the frame, it will conform to the curvature of the frame and fit snugly therein. The finished product shall be symmetrical. The bottom of the lens shall be smooth and free from flash.

* 3.3.6.1 Color (lens).- The color of the lens shall be essentially neutral gray having the following trichromatic characteristics when tested as specified in 4.3.1.1 (see 6.3).

Percent $Y_T$ = 15 percent to 30 percent transmittance
$x$ = 0.270 - 0.350 transmittance
$y$ = 0.300 - 0.380 transmittance

3.3.6.2 Lens covering.- Both sides of the lens of the finished sunglasses shall be covered with smooth finish paper conforming to UU-P-405 to prevent damage to the lens. The lens covering shall be die-cut from one piece to the size and shape of the assembled lens. The design shall be such that, when folded flat on itself, the lens covering will completely cover both front and back of lens and not include the brow bar.

3.3.7 Carrying case.- A carrying case conforming to 3.2.3 shall be furnished with each pair of sunglasses. The case shall approximate color number X34087 of FED-STD-595. The outer surface of the case covering shall have a simulated natural grain leather finish; the inside surface of the case covering shall have a smooth finish.

3.4 Performance.-

3.4.1 Aging.- The sunglasses shall show no discoloration, crazing, cracking, bubbles, surface deterioration of the lens, or separation of the lens from the frame when tested as specified in 4.3.5.



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3.5 Instructions for use.- Instructions covering precautions and care of the lens and sunglasses shall be printed on the outside front of the lens covering specified in 3.3.6.2. The word "Instructions" shall be placed at the top of the lens covering above the nose bridge.

On the left half of the lens covering shall be placed the following wording:

"DO NOT SCRATCH THE LENS. The plastic lens of this sunglass is shatter-resistant, safer and lighter than glass lenses, but easily scratched if abused. When not in use, protect it against dirt or hard objects. To make the lens last longer, clean dust and fingerprints by (1) washing with water, (2) steaming with the breath.

Then wipe dry with a soft clean cloth or lint free paper, making absolutely certain the cloth or paper is free of grit, sand or dirt."

On the right half of the lens covering shall be placed the following wording:

"Gasoline or hot water should not be used for cleaning.

The flexible ear piece of the temple is easily adjusted for proper fit by gently forming with the fingers.

The lens cover should be used at all times when sunglasses are not in use.

Precaution: These sunglasses are to be used for protection against reflected sunlight. DO NOT USE for direct viewing of the sun, welding arcs, etc., or for night driving."

* 3.6 Identification marking.- The carrying case shall be marked as shown in Drawing 2-1-54 with MIL-STD-130 with the following specific information.

U.S.
Sunglasses MIL-S-475D
Manufacturer's Identification
Year of Manufacture
Contract or Order Number

3.7 Workmanship.- The frame shall be clean and free from cracks, weld marks, sink marks and warping; and shall have an accuracy of fit for the lens. The exposed metal parts shall be smoothly finished. The lens shall be clean and free from superficial undulations, lenticulations, striae, bubbles, seeds, scratches, opaque particles or any other imperfections which will impair the visual properties. The finished sunglasses and case shall conform to the levels of quality specified in section 4 herein.



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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Bid sample inspection.- The bid samples specified in 3.1 shall be inspected in accordance with 4.3.3.1.1, 4.3.3.1.2 and 4.3.3.1.3.

* 4.3 Inspection.- Sampling for inspection shall be in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.3.1.1 Testing of components.- In addition to inspections required under referenced specifications, lenses shall be tested in accordance with table I. The sample unit shall be five (5) lenses. The sample size for the designated lot size shall be as follows:

| Lot size (lenses) | Sample size (sample units) |
|---|---|
| 800 or less | 5 |
| 801 up to and including 22,000 | 10 |
| 22,001 and more | 15 |

There shall be no failure of any sample unit to meet the requirements as specified.



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TABLE I.- Component tests

| Component | Requirement paragraph | Test method | Requirements applicable to sample unit | Number of determinations per sample unit | Results reported as | |
|---|---|---|---|---|---|---|
| | | | | | Pass or fail | Numerically to nearest |
| Lens: | | | | | | |
| Luminous transmittance | 3.2.2.1 | Method 3022 of FED-STD-406 | X | 3 | - | 1.0 percent |
| Thickness | 3.3.6 | Micrometer 1/ | X | 1 | - | .001 inch |
| Color | 3.3.6.1 | 4.4.2 | X | 1 | X | - |

1/ The micrometer shall be capable of measuring to 0.0001 inch.





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4.3.1.2 Certificate of compliance.- Components and materials listed below may be accepted on the basis of a supplier's certificate of compliance for characteristics indicated.

| Component | Characteristic | Requirement paragraph |
|---|---|---|
| Plastic (brow bar, nose bridge, temple butt and tubing) | Composition, and percentage of reground plastic (where permitted) | 3.2.1.1, 3.2.1.2, 3.2.1.3 and 3.2.2 |
| Plastic lens material (cellulose acetate butyrate) | Slow burning<br>Composition | 3.2.2.1<br>3.2.2.1 |
| Hinge (temple) | Composition and temper | 3.2.1.4.1 |
| Temple, butt and cable (winding and core) | Composition and temper | 3.2.1.4.2 |
| Rivet (hinge) | Alloy and temper | 3.2.1.4.3 |
| Carrying case (vinyl film covering) | Finish | 3.3.7 |
| Snap fastener | Material and finish | 3.2.3.2 |
| Nose block (wood or plastic, as applicable) | Material and finish (as applicable) | 3.2.3.3 |
| Reinforcement | Material and finish | 3.2.3.4 |

4.3.2 In-process inspection.- Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance. In-process inspection shall be conducted to see that accomplishement of the following is in accordance with the specification requirements:



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| Requirement, operation or assembly | Requirement paragraph |
|---|---|
| Cementing of the temple butt and tubing | 3.3.3 |
| Hinge is spot welded, soldered or riveted to the temple butt insert | 3.3.4 |
| Brow bar and nose bridge cemented to lens | 3.3.5 |
| Cable consists of 2 or more strands of wire wound around wire core and reduced to 0.070 ± 0.002 inch diameter | Drawing 2-1-54 |

4.3.3 Inspection of the end item.-

4.3.3.1 Examination of sunglasses with carrying case.- The defects found during visual examination shall be classified in accordance with 4.3.3.1.1, 4.3.3.1.2 and 4.3.3.1.3. The sample unit for these examinations shall be one pair of sunglasses with case. The lot size shall be the number of end items in the inspection lot. The inspection levels and the acceptable quality levels (AQLs) for end item examinations shall be as listed in 4.3.3.1.4.

4.3.3.1.1 Visual examination.-

NOTE: Defects designated with an asterisk (*) shall be classified as major when seriously affecting serviceability or appearance, and minor when affecting serviceability or appearance but not seriously.

| Examine | Defect | Classification Major | (*) | Minor |
|---|---|---|---|---|
| Sunglasses: | | | | |
| Quality of plastic material | Break, crack or fracture | X | | |
| | Crazing, weld mark, scratch, fissure, or otherwise impaired | | (*) | |
| | Brow bar, nose bridge or temple butt not opaque black color | | | X |
| | Color of brow bar and nose bridge non-uniform, off-shade, mottled or streaky | | | X |



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| Examine | Defect | Classification Major | (*) | Minor |
|---|---|---|---|---|
| Sunglasses: (cont'd) | | | | |
| Quality of plastic material (cont'd) | Surface of brow bar, nose bridge, temple butt or temple tubing not smooth or not uniform | | (*) | |
| | Lens not optically clear | X | | |
| | Temple tubing not clear | | | X |
| | Lens has objectionable roughness | X | | |
| | Lens contains superficial undulation, lenticulation or striae, which affects visual properties | X | | |
| | Lens not high polish finish | X | | |
| | Lens contains stain or discoloration not readily removed with water or steaming with breath, then wiping dry with soft paper | X | | |
| | Lens not symmetrical | X | | |
| | Lens shows bubbles or surface deterioration | X | | |
| | Lens surface abraded or dulled | X | | |
| | Edge of lens cracked, rough, not smooth finish, or has any flash | X | | |
| Quality of hardware components | Not fabricated of brass or nickel silver as specified | X | | |
| | Component fractured, split, dented, bowed, sprung, slivered, broken, chipped, bent out of shape, or otherwise impaired | | (*) | |
| Quality of lens covering | Not paper | | | X |
| | Has rough finish which may damage lens | X | | |
| | Any sliver, shive, dirt spot, grease or foreign matter | | | X |



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| Examine | Defect | Classification Major | (*) | Minor |
|---|---|---|---|---|
| Carrying case: | | | | |
| Quality of covering | Does not approximate required color | | | X |
| | Color not uniform; stained or discolored | | | X |
| | Outer surface does not have simulated natural grain leather finish | | | X |
| | Inside surface does not have smooth finish | | (*) | |
| | Torn, repaired, patched, cracked or otherwise impaired | X | | |
| Quality of hardware (snap fastener and reinforcement) | Not fabricated of brass or steel as specified | X | | |
| | Component fractured, split, bent out of shape, dented, bowed, sprung, slivered, burred, broken, chipped, malformed or otherwise impaired | | (*) | |
| | Snap fastener does not consist of screw stud or stud and eyelet (as applicable), socket and button | X | | |
| | Snap fastener not specified size, i.e. not 16 line | | | X |
| | Snap fastener does not have black chemical finish | | | X |
| | Black chemical finish flakes or rubs off | | | X |
| | Evidence of rust or corrosion | X | | |
| Quality of wood block (as applicable) | Not wood | | | X |
| | Check, shake or split, loose or unsound knot, honeycomb or decay which may impair use | | | X |
| | Wood chipped or splintered | | | X |
| | Pitch pocket or mineral streak | | | X |
| | Surface not smooth, contains sharp or rough edge | X | | |
| | No evidence of sealer | | (*) | |
| | Surface tacky | | (*) | |



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| Examine | Defect | Classification Major | (*) | Minor |
|---|---|---|---|---|
| Carrying case: (cont'd) | | | | |
| Quality of plastic block (as applicable) | Surface not smooth | | | X |
| | Sharp or rough edge | X | | |
| | Warped, void, blister or break | | | X |
| Design and construction | Varies from bid sample or specified drawing, as applicable | X | | |
| | Any characteristic not in accordance with the specified design, e.g., not designed to accommodate plastic lens and provide binocular vision | X | | |
| Workmanship and assembly of sunglasses | Any component missing | X | | |
| | Component malformed, warped, chipped or otherwise damaged | | (*) | |
| | The vertical and horizontal angles of the two temples are not matched and do not provide a satisfactory fit at the mitered joint with the brow bar | | (*) | |
| | Mating barrels of the hinge loose, e.g., permits angular up and down movement of the temple | | (*) | |
| | Evidence of burn or discoloration on hinge | | | X |
| | Not connected or not joined as specified or assembly is poorly accomplished | X | | |
| | Rivet loose, i.e., not permanently or not securely set | X | | |
| | Rivet damaged in setting | X | | |
| | Rivet inverted, e.g., head not on top when sunglasses are in wearing position | | | X |
| | Tubing loose in temple butt, i.e., not securely cemented together | X | | |



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| Examine | Defect | Classification Major | (*) | Minor |
|---|---|---|---|---|
| Workmanship and assembly of sunglasses (cont'd) | Lens not attached as specified, e.g., not cemented in brow bar, not in brow bar lens groove, etc. | X | | |
| | Lens does not conform to curvature of frame, does not fit snugly therein | X | | |
| | Lens not centered on brow bar | X | | |
| | Any area of no bond between lens and brow bar | | (*) | |
| | Lens set in brow bar crooked, at an angle or not fully set in brow bar | | (*) | |
| | Nose bridge loose, i.e., not securely cemented to lens | X | | |
| | Nose bridge not properly positioned e.g., crooked, set at angle, set in a reversed position, etc. | X | | |
| | Spots, drippings runs or splatters of cement on brow bar, nose bridge, temple or lens | X | | |
| | Core winding is irregular or wound on core loosely | | | X |
| | Core winding does not extend from temple butt to the heat sealed end of tubing | | | X |
| | Any edge not trimmed | | (*) | |
| Workmanship and assembly of carrying case | Case not heat sealed as shown on drawing | X | | |
| | Any seam area in case that has not been sealed | X | | |
| | Poor heat seal resulting in inadequate fusion | X | | |
| | Snap fastener component missing or not type specified | X | | |
| | Snap fastener set excessively tight, cutting surrounding plastic | X | | |



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| Examine | Defect | Classification Major | (*) | Minor |
|---|---|---|---|---|
| Workmanship and assembly of carrying case (cont'd) | Snap fastener loose, not securely set or retained, or not properly assembled | X | | |
| | Male and female parts of snap fastener on case cannot be closed, or cannot be separated from closed positon, ineffective e.g., will not engage socket or otherwise defective | X | | |
| | Snap fastener requires undue force to operate but will function | | | X |
| | Carrying case not connected or joined as specified or assembly is poorly accomplished | X | | |
| Identification marking and instruction sheet (lens covering) | Identification marking missing, incomplete, illegible, misspelled, not specified size or type, not in specified locations on carrying case, or not of permanent nature | | | X |
| | Instruction sheet missing | X | | |
| | Cut, hole, tear or break in instruction sheet | | | X |
| | Instruction sheet not in one piece, not shaped to completely cover both sides of lens, or is not removable | | | X |
| | Instructions vary from specified wording or wording is not placed as specified | | | X |

4.3.3.1.2 Dimensional examination.- Any dimension that is not within the specified tolerance shall be classified as a defect.

4.3.3.1.3 Examination of sunglasses for fit and vision.-



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| Examine | Defect |
|---|---|
| Fit and vision | Flexible ear piece of the temple is too rigid or too flexible to be easily adjusted for proper fit by gently forming with the fingers<br>Flexible ear piece when adjusted for fit will not retain fit<br>Vision has area of distortion or disrupted visual acuity |

4.3.3.1.4 Inspection levels and AQLs.- The inspection levels for determining the sample size and the AQLs expressed in defects per 100 units shall be as follows:

| | Inspection level | AQL | |
|---|---|---|---|
| | | Major | Total |
| For defects applicable to 4.3.3.1.1 | II | 6.5 | 15.0 |
| For defects applicable to 4.3.3.1.2 | S-3 | One class | 4.0 |
| For defects applicable to 4.3.3.1.3 | S-3 | One class | 4.0 |

4.3.4 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing and marking comply with the requirements of section 5 of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

| Examine | Defect |
|---|---|
| Marking (exterior and interior) | Omitted; incorrect; illegible; of improper size, location sequence or method of application.<br>Marking on end item does not correspond to marking on interior package or shipping container |
| Materials | Any component missing<br>Any component damaged, affecting serviceability |



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| Examine | Defect |
|---|---|
| Workmanship | Inadequate application of components, such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulged or distorted container. |
| Content | Number of interior packages is more or less than specified, number of end item units per interior package is more or less than specified. 1/ |

1/ In examination of number of end items per package, two (2) packages shall be examined from each shipping container in the sample.

4.3.4.1 Examination for palletization.- An examination shall be made to determine that the palletization complies with the section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

| Examine | Defect |
|---|---|
| Finished dimensions | Length, width, or height exceeds specified maximum requirement. |
| Palletization | Not as specified.<br>Pallet pattern not as specified.<br>Interlocking of loads not as specified.<br>Load not bonded with required straps as specified. |
| Weight | Exceeds maximum load limits. |
| Marking | Omitted; incorrect; illegible; of improper size, location, sequence or method of application. |

* 4.3.5 Testing of the end item.- The finished sunglasses shall be tested as specified in 4.4.1 to determine conformance with the requirements in 3.4.1. The lot size shall be expressed in units of one pair of sunglasses each. The sample unit shall be one pair of sunglasses. The sample size for testing shall be in accordance with 4.3.1.1. There shall be no failure of any sample to meet the requirements as specified. All test reports shall contain the individual values utilized in expressing the final result.



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4.4 Test methods.-

4.4.1 Aging.- The finished sunglasses shall be placed in a circulating air oven at a temperature of 150°F ± 2°F for 24 hours, allowed 1 hour at room temperature to return to normal and then subjected to a reduced temperature of minus 10°F ± 2°F for a minimum of 12 hours. The sunglasses shall then be examined for presence of discoloration, crazing, cracking, bubbles, surface deterioration of the lens or separation of the lens from the frame.

4.4.2 Color.- The lens shall be inspected for color by use of a spectrophotometer measuring against standard source "C" with magnesium oxide as 100 percent reference (see 6.3).

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

* 5.1.1 Level A.- Twenty sunglasses, each in a closed carrying case, shall be packaged on edge 2 in length, 10 in width, and 1 in depth within a folding paperboard box conforming to variety 1, style III, type G, class 1 of PPP-B-566; setup paperboard box conforming to type I, variety 1, class A, style 4 of PPP-B-676; or metal-stayed paperboard box conforming to style A of PPP-B-665. Each interior box shall be provided with a full width and depth divider pad made of the same material as the box. Inside dimensions of each paperboard box shall be 12-3/4 inches in length, 9-1/8 inches in width, and 2-7/8 inches in depth. Each box shall be securely closed with 2 inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45, applied at the center of the length opening along the bottom and up each side a minimum of 1-1/2 inches.

5.1.2 Level C.- Sunglasses (with case) shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A.- Three hundred sunglasses (with cases), packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A packages shall be packed on end, three in length, five in width, and one in depth within a shipping container. Inside dimensions of each shipping container shall approximate 28-1/2 inches in length, 15-1/4 inches in width, and 13 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed, waterproofed and reinforced in accordance with the appendix of the container specification.



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* 5.2.2 Level B.- Three hundred sunglasses (with cases), packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A packages shall be packed on end, three in length, five in width, and one in depth within a shipping container. Inside dimensions of each shipping container shall approximate 28-1/2 inches in length, 15-1/4 inches in width, and 13 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

* 5.2.2.1 When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the container specification.

* 5.2.3 Level C.- Sunglasses (with cases), packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

* 5.3 Palletization.- Unless otherwise specified (see 6.2), sunglasses (with cases), packed as specified in 5.2, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

* 5.4 Marking.- In addition to any special marking required by the contract or order, interior packages, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129.

## 6. NOTES

6.1 Intended use.- The sunglasses with case are intended for use by military personnel for protection against excessive light and glare from reflected sunlight.

* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When the bidder intends to furnish an alternate design (see 3.1), approval of bid sample should be obtained from U. S. Army Natick Laboratories, Natick, Mass. 01760, ATTN: Clothing and Personal Life Support Equipment Laboratory.



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(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(e) When palletization is not required (see 5.3).

6.3 The colorimetric characteristics specified in 3.3.6.1 encompass essentially the range of Munsell values falling between 5B 4/2, 5G 4/2 and 5Y 4/2.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 11

Preparing activity:

Army - GL

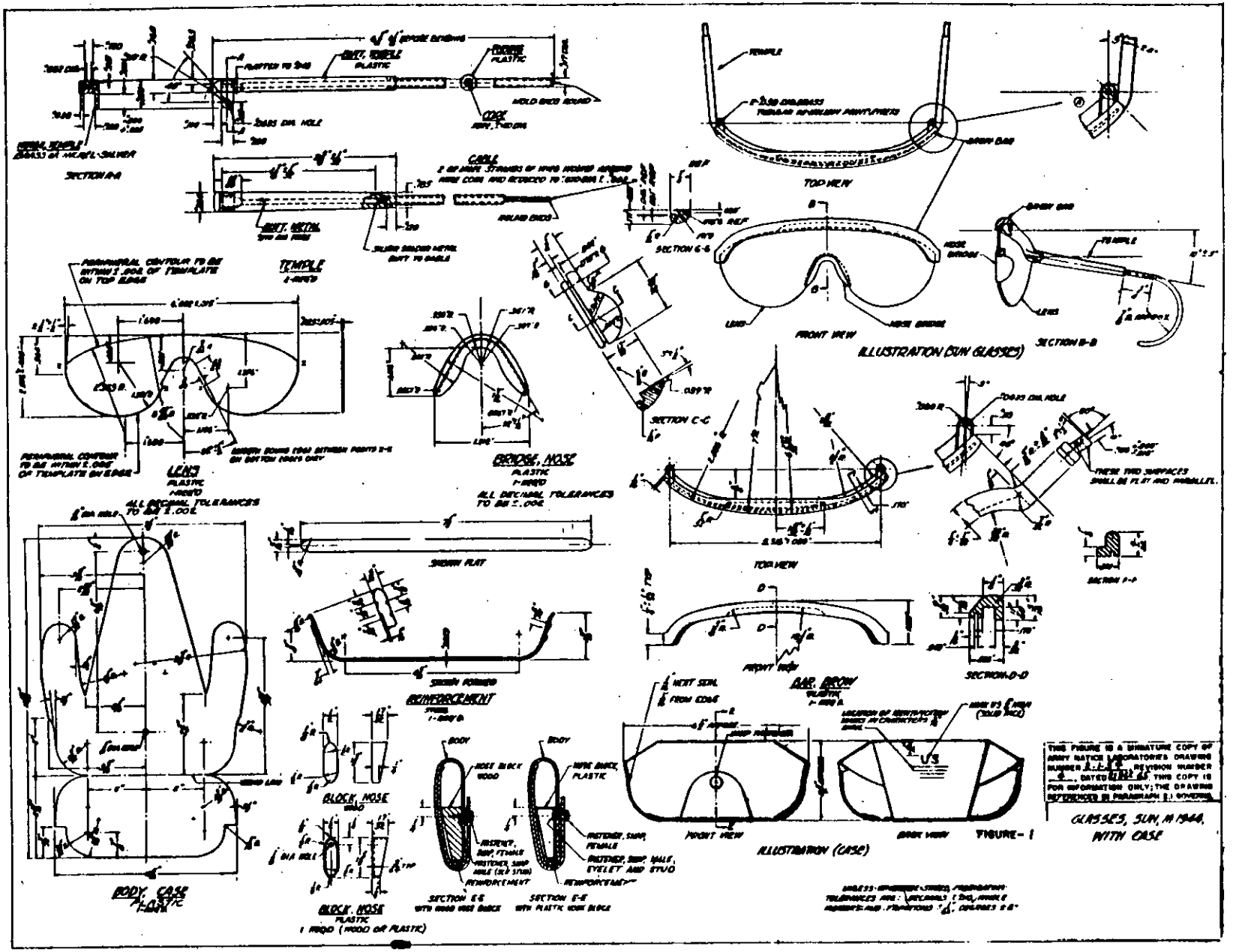
Project No. 8465-0288

Review activities:

Army - MD
Navy - AS, MC, MS
Air Force - 82



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

**5. PROBLEM AREAS**

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

**6. REMARKS**

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |
| | |

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